

D789

# Applied Machine Learning for Business Solutions

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

A telecom wants to predict whether each customer will cancel in the next 90 days. It has three years of history labeled “churned” or “stayed,” stored in a cloud warehouse bought last year. What type of ML problem is this?

- A. Supervised binary classification.
- B. Whatever method the cloud warehouse supports best.
- C. Regression.
- D. Clustering.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2.

A fraud model flagged 200 transactions, and 150 of them were actually fraud. There were 250 fraudulent transactions among 100,000 in total. The model was built by an outside consultancy. What are its precision and recall?

- A. They cannot be found without the consultancy’s report.
- B. Precision 60%, recall 75%.
- C. Precision 75%, recall 60%.
- D. Precision 75%, recall 0.15%.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3. [TRANSFER — reversed stem / different variable]

A hospital screening model must catch nearly every true case, and follow-up tests for false alarms are cheap. Which metric should the team prioritize?

- A. Precision.
- B. Overall accuracy.
- C. Precision, because only precision matters in healthcare.
- D. Recall.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

About 2% of loan applications are fraudulent. A vendor with 400 clients says its model is 98% accurate. What should the analyst conclude?

- A. 98% could come from labeling every application legitimate; check precision and recall.
- B. The model catches 98% of fraud.
- C. The model is excellent.
- D. Trust it because the vendor has many clients.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5. [TRANSFER — reversed stem / different variable]

Of 10,000 emails, 500 are spam. What accuracy does a model get if it labels every email “not spam”?

- A. 0%
- B. 95%
- C. 5%
- D. 50%

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6.

A model scores 99% accuracy on training data and 71% on validation data. The team trained it on a new GPU cluster over a weekend. What is happening, and what helps?

- A. Underfitting; make the model more complex.
- B. Train it longer on the same data.
- C. Use faster GPUs.
- D. Overfitting; simplify or regularize the model or add training data.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7. [TRANSFER — reversed stem / different variable]

A model scores 62% on training data and 60% on validation, while a simple benchmark scores 80%. What is likely, and what helps?

- A. Always collect more data first.
- B. Overfitting; add regularization.
- C. Underfitting; add useful features or use a more capable model.
- D. It is fine because the two scores are close.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8.

A model predicting 30-day hospital readmission uses a feature called “follow-up call outcome,” recorded two weeks after discharge. Validation AUC is 0.99. The hospital has 600 beds. What should the team do?

- A. Deploy it; 0.99 is excellent.
- B. Remove features not available at prediction time and re-validate.
- C. Collect data from more beds.
- D. Add more post-discharge features to push AUC higher.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9.

A retention offer costs \$20 per customer targeted, and each saved customer is worth \$300. The model flags 1,000 customers; 10% of them would actually churn, and the offer saves half of those. The marketing VP prefers email campaigns. What is the net value of the campaign?

- A. -\$20,000
- B. -\$5,000
- C. +\$15,000
- D. +\$10,000

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

10. [TRANSFER — reversed stem / different variable]

A retention offer costs \$20 per customer targeted, and each saved customer is worth \$300. What share of targeted customers must be saved to break even?

- A. About 6.7%
- B. 15%
- C. 20%
- D. About 93.3%

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**11.**

A team tuned its model 40 times, checking the test set each time and keeping the best test score of 91%. The data came from a vendor in Germany. What is the problem?

- A.** Nothing; the best test score is the expected real performance.
- B.** Tune on the training set's score instead.
- C.** The 91% is too optimistic; tune on validation data and use the test set once.
- D.** The vendor's location biases the score.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**12.**

A data scientist will present a churn model to the executive team. Her slides open with gradient-boosting settings and ROC curves. The meeting is 15 minutes on a Friday afternoon. What should she lead with instead?

- A.** A request to move the meeting to Monday.
- B.** Detailed model settings, to build technical credibility.
- C.** A promise that the model is 100% reliable.
- D.** The recommendation, its expected business impact, and the key uncertainties.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer A Concept 1

A telecom wants to predict whether each customer will cancel in the next 90 days. It has three years of history labeled “churned” or “stayed,” ~~stored in a cloud warehouse bought last year~~. What type of ML problem is this?

**Target word:** problem type **Decides:** A yes/no label exists for past customers.

|   |                   |                                                            |
|---|-------------------|------------------------------------------------------------|
| A | <b>Correct</b>    | Labeled data and a two-category outcome.                   |
| B | Camouflage        | The storage platform does not define the problem.          |
| C | Confusion set     | Regression predicts a number, not a category.              |
| D | Adjacent category | Clustering finds groups without labels; labels exist here. |

## 2. Answer C Concept 2

A fraud model flagged 200 transactions, and 150 of them were actually fraud. There were 250 fraudulent transactions among 100,000 in total. ~~The model was built by an outside consultancy~~. What are its precision and recall?

**Target word:** precision and recall **Decides:** Precision uses the 200 flagged; recall uses the 250 actual.

|   |                    |                                                              |
|---|--------------------|--------------------------------------------------------------|
| A | Camouflage         | The stem gives every count needed.                           |
| B | Inverted operation | Swaps the two denominators.                                  |
| C | <b>Correct</b>     | $150 / 200 = 75\%$ ; $150 / 250 = 60\%$ .                    |
| D | Wrong stem number  | Divides by all 100,000 transactions instead of actual fraud. |

## 3. Answer D Concept 2 · transfer

A hospital screening model must catch nearly every true case, and follow-up tests for false alarms are cheap. Which metric should the team prioritize?

**Target word:** choose the metric from costs **Decides:** Misses are costly; false alarms are cheap.

|   |                    |                                                      |
|---|--------------------|------------------------------------------------------|
| A | Inverted operation | Precision limits false alarms, which are cheap here. |
| B | Adjacent category  | Accuracy can stay high while missing rare cases.     |
| C | Absolute language  | The right metric depends on the costs of each error. |
| D | <b>Correct</b>     | Recall measures how many true cases are caught.      |

## 4. Answer A Concept 3

About 2% of loan applications are fraudulent. A vendor ~~with 400 clients~~ says its model is 98% accurate. What should the analyst conclude?

**Target word:** interpret accuracy on rare events **Decides:** 98% equals the non-fraud share.

|   |                 |                                                            |
|---|-----------------|------------------------------------------------------------|
| A | <b>Correct</b>  | Accuracy matches the majority share, so it proves nothing. |
| B | Confusion set   | Confuses accuracy with recall.                             |
| C | Dropped element | Ignores the class imbalance.                               |
| D | Camouflage      | Client count is not evidence of performance.               |

### 5. Answer B Concept 3 · transfer

Of 10,000 emails, 500 are spam. What accuracy does a model get if it labels every email “not spam”?

**Target word:** accuracy of the majority-class guess **Decides:** 9,500 correct out of 10,000.

|          |                    |                                                      |
|----------|--------------------|------------------------------------------------------|
| A        | Confusion set      | That is its recall on spam, not accuracy.            |
| <b>B</b> | <b>Correct</b>     | $9,500 / 10,000 = 95\%$ .                            |
| C        | Inverted operation | That is the spam share it gets wrong.                |
| D        | Dropped element    | Ignores the class counts, as if it were a coin flip. |

### 6. Answer D Concept 4

A model scores 99% accuracy on training data and 71% on validation data. The team trained it ~~on a new GPU cluster over a weekend~~. What is happening, and what helps?

**Target word:** diagnose and treat **Decides:** Large train-validation gap.

|          |                    |                                                            |
|----------|--------------------|------------------------------------------------------------|
| A        | Inverted operation | Adds complexity, widening the gap.                         |
| B        | Dropped element    | More training usually deepens overfitting.                 |
| C        | Camouflage         | Hardware speed does not change generalization.             |
| <b>D</b> | <b>Correct</b>     | The model memorized training data and does not generalize. |

### 7. Answer C Concept 4 · transfer

A model scores 62% on training data and 60% on validation, while a simple benchmark scores 80%. What is likely, and what helps?

**Target word:** diagnose low scores on both sets **Decides:** Low training score.

|          |                    |                                                                 |
|----------|--------------------|-----------------------------------------------------------------|
| A        | Absolute language  | More data rarely fixes a model too simple to learn the pattern. |
| B        | Inverted operation | Regularization makes an underfit model weaker.                  |
| <b>C</b> | <b>Correct</b>     | It cannot even fit the training data.                           |
| D        | Dropped element    | Ignores that both scores are poor.                              |

### 8. Answer B Concept 5

A model predicting 30-day hospital readmission uses a feature called “follow-up call outcome,” recorded two weeks after discharge. Validation AUC is 0.99. ~~The hospital has 600 beds.~~ What should the team do?

**Target word:** respond to a suspicious score **Decides:** The feature is recorded after prediction time.

|          |                    |                                                                |
|----------|--------------------|----------------------------------------------------------------|
| A        | Dropped element    | Ignores when the feature becomes known.                        |
| <b>B</b> | <b>Correct</b>     | The feature leaks the outcome; real performance will be lower. |
| C        | Camouflage         | Hospital size does not fix leakage.                            |
| D        | Inverted operation | Adds more leakage.                                             |

### 9. Answer B Concept 6

A retention offer costs \$20 per customer targeted, and each saved customer is worth \$300. The model flags 1,000 customers; 10% of them would actually churn, and the offer saves half of those. ~~The marketing VP prefers email campaigns.~~ What is the net value of the campaign?

**Target word:** net value **Decides:** 50 customers saved against 1,000 offers.

|   |                   |                                                                  |
|---|-------------------|------------------------------------------------------------------|
| A | Wrong stem number | The offer cost alone.                                            |
| B | <b>Correct</b>    | $50 \times \$300 = \$15,000$ benefit – \$20,000 cost = –\$5,000. |
| C | Confusion set     | Gross benefit without subtracting the offer cost.                |
| D | Dropped element   | Drops the 50% save rate: $100 \times \$300 - \$20,000$ .         |

### 10. Answer A Concept 6 · transfer

A retention offer costs \$20 per customer targeted, and each saved customer is worth \$300. What share of targeted customers must be saved to break even?

**Target word:** solve for break-even save rate **Decides:** Save rate  $\times$  \$300 = \$20.

|   |                    |                                                  |
|---|--------------------|--------------------------------------------------|
| A | <b>Correct</b>     | $\$20 / \$300 = 0.0667$ .                        |
| B | Inverted operation | Divides \$300 by \$20 and reads it as a percent. |
| C | Wrong stem number  | Uses the \$20 cost as a percent.                 |
| D | Confusion set      | The share not saved at break-even.               |

### 11. Answer C Concept 7

A team tuned its model 40 times, checking the test set each time and keeping the best test score of 91%. ~~The data came from a vendor in Germany.~~ What is the problem?

**Target word:** flaw in evaluation **Decides:** The test set was used for tuning.

|   |                   |                                                      |
|---|-------------------|------------------------------------------------------|
| A | Dropped element   | Ignores selection bias from 40 checks.               |
| B | Adjacent category | Training scores are even more optimistic.            |
| C | <b>Correct</b>    | Repeated test checks let the model fit the test set. |
| D | Camouflage        | The data source is not the flaw here.                |

### 12. Answer D Concept 8

A data scientist will present a churn model to the executive team. Her slides open with gradient-boosting settings and ROC curves. The meeting is 15 minutes ~~on a Friday afternoon~~. What should she lead with instead?

**Target word:** what to lead with **Decides:** Executive audience, short time.

|   |                   |                                                     |
|---|-------------------|-----------------------------------------------------|
| A | Camouflage        | The day does not change what to lead with.          |
| B | Confusion set     | Fits a technical review, not an executive decision. |
| C | Absolute language | Overstates certainty and misleads decision makers.  |
| D | <b>Correct</b>    | Executives need decisions and value in their terms. |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 7             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 3             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 3             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 7             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 3             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 6             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 7             |             |

## Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                 | Rule                                                                                                                                                                     | Trap                                                                           | Tell                                                     |
|---|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------|
| 1 | <b>Framing the ML problem</b>           | Labeled categories mean classification; a labeled number means regression; no labels and a need to find groups means clustering.                                         | Choosing the method from the data platform or buzzwords instead of the target. | What the prediction looks like and whether labels exist. |
| 2 | <b>Precision vs. recall</b>             | Precision = true positives / predicted positives; recall = true positives / actual positives. Prioritize recall when misses are costly, precision when false alarms are. | Swapping the denominators.                                                     | Counts of flagged items and of actual cases.             |
| 3 | <b>Accuracy with class imbalance</b>    | When one class is rare, a model that always predicts the common class scores high accuracy while finding nothing.                                                        | Taking high accuracy as proof of a good model.                                 | A rare event and an accuracy near the majority share.    |
| 4 | <b>Overfitting vs. underfitting</b>     | High training and much lower validation scores mean overfitting; low scores on both mean underfitting.                                                                   | Applying the fix for one to the other.                                         | Training and validation scores side by side.             |
| 5 | <b>Data leakage</b>                     | A feature that would not be known at prediction time makes validation scores look better than real use.                                                                  | Celebrating a near-perfect score.                                              | A feature recorded after the event being predicted.      |
| 6 | <b>Business value of a model</b>        | Judge a model by the net value of acting on its predictions: benefit of correct actions minus the cost of all actions taken.                                             | Counting gross benefit or ignoring how many actions actually succeed.          | A cost per action and a value per success.               |
| 7 | <b>Train, validation, and test sets</b> | Tune on validation data; use the test set once for an unbiased final estimate.                                                                                           | Tuning against the test set and reporting the best score.                      | Repeated checks of the test score during tuning.         |
| 8 | <b>Communicating to stakeholders</b>    | Lead with the decision, business impact, and key uncertainties in the audience's terms; keep technical detail for those who need it.                                     | Leading with algorithms, or overstating certainty.                             | An executive audience and a short time slot.             |

**Course scope (public description):** This course is designed to familiarize students with machine learning through the lens of a software engineer. Students will explore the challenges and opportunities of applying machine learning to solve problems and create strategic objectives. Students will explore various industries and learn how to apply machine learning to address business needs. While in this course, students will demonstrate how to effectively communicate their recommendations to a range of stakeholders.